

Electrification of Passenger Cars in Austria

Vienna in comparison with the Austrian federal states, 2004–2023

Executive Summary

Objective. This project examines how the number and composition of electric and hybrid passenger cars changed in Austria and Vienna between 2004 and 2023, with a focus on four questions: long-term development, relative growth, the changing Electric Vehicle (EV) / Hybrid mix, and Vienna's position compared with other federal states.

Key findings. Both electric and hybrid vehicles increased substantially over the observed period. In Vienna, electric vehicles accelerated particularly strongly after the late 2010s. Although hybrids remained more numerous, EV growth was faster in recent years, increasing the share of EVs within the EV+Hybrid group to about 35% by 2023. Vienna ranked third in absolute EV count in 2023, but its EV share within the EV+Hybrid group was among the lowest of the nine federal states.

1. Business Questions

1. How did the number of electric and hybrid passenger cars change over time?
2. Which category grew faster: electric vehicles or hybrid vehicles?
3. How did the relationship between EVs and hybrids change over time?
4. How did Vienna compare with the other Austrian federal states in 2023?

2. Data and Scope

Source. Official Statistik Austria open data on passenger cars with electric or hybrid drive. The 2023 dataset contains observations for Austrian federal states and a historical year field covering 2004–2023.

Core variables used. YEAR, FEDERAL_STATE, ELECTRICITY, and HYBRID. Unused source fields were removed during preparation.

Important data note. Hybrid values are missing for 2004–2005. Because this missingness also appears in the earlier version of the source, the analysis treats 2006 as the first year in which EV and Hybrid can be compared directly. The original HYBRID field is preserved as null for 2004–2005.

3. Preparation and Analysis

Stage	What was done
Data preparation	The raw CSV was imported into Excel and prepared with Power Query. Only the four analytical fields required for the project were retained, data types were checked, and the source query was preserved as a reference copy.
Derived measures	TOTAL_ELECTRIFIED was calculated as ELECTRICITY + HYBRID. EV_SHARE_OF_ELECTRIFIED was calculated as ELECTRICITY / (ELECTRICITY + HYBRID) for 2006 onward.

Analysis	Excel PivotTables were used for descriptive comparisons and year-over-year growth checks. A Power BI DAX measure was then used for the EV-share calculation and interactive visualization.
Interpretation	Absolute vehicle counts and relative shares were analysed together. The project does not interpret EV share as the share of all passenger cars, because the dataset used here does not provide the total passenger-car fleet denominator.

4. Key Findings

1. Long-term growth. The number of both electric and hybrid passenger cars increased strongly over time. In Vienna, EVs increased from 20 in 2006 to 27,745 in 2023, while hybrids increased from 162 to 52,268 over the same period.

2. EV growth accelerated. Although hybrids remained the larger category, EVs showed stronger year-over-year growth in the later part of the period. From 2019 to 2023, EV growth was approximately +71%, +62%, +86%, +54% and +56%, compared with approximately +39%, +34%, +50%, +51% and +36% for hybrids.

3. The mix shifted toward EVs. The EV share within the combined EV+Hybrid group in Vienna increased from around 2–7% during the early 2010s to 19% in 2018, 24% in 2020, 31% in 2022 and 35% in 2023.

4. Vienna is large, but not a leader by EV share. In 2023 Vienna had 27,745 EVs, the third-highest absolute count among Austrian federal states. However, its EV share within EV+Hybrid was about 35%, compared with 43% in Oberösterreich and 42% in Vorarlberg.

5. Vienna vs. Federal States in 2023

Federal state	Electric	Hybrid	EV share of EV+Hybrid
Oberösterreich	30,291	40,680	43%
Vorarlberg	8,834	11,964	42%
Salzburg	12,220	18,356	40%
Burgenland	4,724	7,408	39%
Niederösterreich	30,073	48,932	38%
Steiermark	19,797	32,996	37%
Tirol	14,162	23,877	37%
Wien	27,745	52,268	35%
Kärnten	7,644	14,501	35%

6. Recommendations

1. Continue monitoring the EV transition. The faster growth of EVs in recent years suggests that future reporting should track EV growth and EV share separately rather than combining EV and Hybrid into one category.

2. Use both absolute and relative indicators. Vienna illustrates why a single KPI can be misleading: it has the third-highest absolute EV count but one of the lowest EV shares within EV+Hybrid. Recurring reports should therefore show both counts and shares.

3. Extend the analysis with the total passenger-car fleet. A future version should join the dataset with total passenger-car stock by federal state and year. This would allow calculation of EV penetration across the full passenger-car fleet, which is a stronger measure than the EV share used in this project.

7. Limitations

- Hybrid data are unavailable for 2004–2005, so direct EV-versus-Hybrid comparisons start in 2006.
- The EV share metric is the share of EVs within the combined EV+Hybrid group; it is not the share of EVs in the total passenger-car fleet.
- The analysis is descriptive and identifies patterns; it does not establish causal explanations for differences between federal states.
- The latest year contained in the dataset is 2023.

8. Tools and Deliverables

The project was completed using Excel and Power Query for data preparation and validation, and Power BI for the final dashboard and interactive visualization. The dashboard contains three main views: the development of electric and hybrid vehicles in Vienna, the changing EV share within EV+Hybrid, and the 2023 comparison across Austrian federal states.